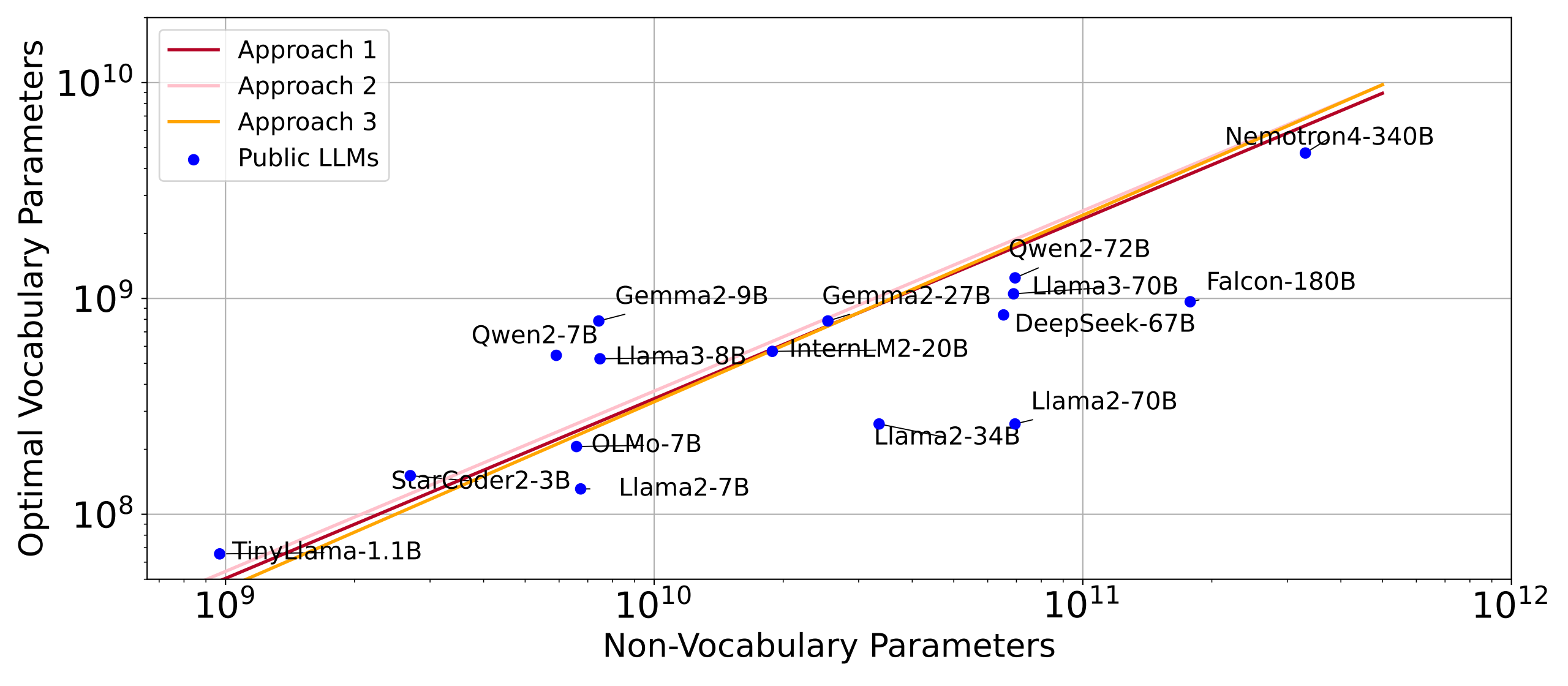
Approach 1
Approach 2
Approach 3
Public LLMs
Optimal Vocabulary Parameters
Non-Vocabulary Parameters
10^{8}
10^{9}
10^{10}
10^{9}
10^{10}
10^{11}
10^{12}
Nemotron4-340B
Qwen2-72B
Llama3-70B
Falcon-180B
DeepSeek-67B
Gemma2-9B
Gemma2-27B
Qwen2-7B
Llama3-8B
InternLM2-20B
Llama2-70B
Llama2-34B
OLMo-7B
StarCoder2-3B
Llama2-7B
TinyLlama-1.1B